Expert Systems With Applications 213 (2023) 118871

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Braess Paradox under the bi-objective user equilibrium

Wenlong Zhu [a], Junting Zhang [a], Shunqiang Ye [a,*], Wanli Xiang [b]

[a] School of Management, Anhui University, Hefei 230601, China
[b] School of Traffic & Transportation Engineering, Lanzhou Jiaotong University, Lanzhou 730070, China

ARTICLE INFO

Keywords:
Braess Paradox
Bi-objective user equilibrium
Route choice

ABSTRACT

Braess Paradox and its derivative paradoxes under the single objective decision-making assumption have been extensively studied. When the single objective decision-making assumption is replaced by bi-objective decision-making assumption, the occurrence conditions under which the Braess Paradox are unknown. The paper investigates Braess Paradox under the bi-objective user equilibrium. Firstly, the definitions of Braess Paradox and Weak Braess Paradox under the bi-objective user equilibrium are introduced. Secondly, the analytical properties of the bi-objective user equilibrium solutions and the occurrence conditions of the Braess Paradox are explored on the classical Braess network with linear link performance functions. It is proved that the occurrence conditions of Braess Paradox are dependent upon link performance parameters and travel demand on the Braess network. Finally, a numerical example is tested to demonstrate Braess Paradox under the bi-objective user equilibrium.

1. Introduction

Constructing new road on existing road network is usually regarded as an effective way to improve the performance of road network, such as decrease the total travel time of network, reduce emissions from vehicles. However, based on the assumption of rational travel behavior, Braess (1968) found that adding a new link to the network with certain link performance parameters may increase the total travel time of network (also see Braess, Nagurney, & Wakolbingger, 2005). This counterintuitive phenomenon that improving road network, including adding new links, or enlarging link capacity of existing network, makes network system performance worse is called as Braess Paradox. In actual transportation planning, road planners should consider and avoid the deterioration of network performance caused by Braess Paradox.

After this discovery, Braess Paradox and its variants have been concerned and investigated by many scholars, and have been extensively applied to airway networks (Ma et al., 2019), multi-retailer systems (Çömez-Dolgan & Fescioglu-Unver, 2015), evolutionary variational inequality model of the Internet (Nagurney, Parkes, & Daniele, 2007), large random graphs (Valiant & Roughgarden, 2006) and other fields. Murchland (1970) showed that the Braess Paradox occurred on the major road network of the city center in Stuttgart. Steinberg and Zangwill (1983) given the efficient and necessary conditions for the occurrence of Braess Paradox. Nagurney (2010) proved that the Braess Paradox would disappear under higher demands in congested networks. Bagloee et al. (2014) developed a heuristic methodology based on a Genetic Algorithm to tackle Braess Paradox detection problem. Different from Braess Paradox in the framework of non-cooperative game-theoretic, Passacantando et al. (2021a) first studied Braess Paradox in a cooperative game-theoretic setting. Subsequently, Passacantando et al. (2021b) investigated Braess Paradox and average quality of service in a cooperative game-theoretic setting. Besides, some scholars adopted experimental research to verify the existence of Braess Paradox in different scenarios. Rapoport et al. (2008) designed traffic network game experiment to test the Braess Paradox, and showed the Braess Paradox also occurs in more complex and asymmetric network settings. Morgan, Orzen and Sefton (2009) found that technical improvements of road network reduced users' benefits on the route choice experiment. Liu et al. (2020) designed the laboratory-like day-to-day route choice experiment to examine the Braess Paradox under the condition of travelers equipped with Advanced Traveler Information Systems (ATIS), and found that all travelers with traffic information cannot weaken Braess Paradox. Bittihn and Schadschneider (2020) found that modern traffic information systems cannot overcome Braess Paradox. Bittihn and Schadschneider (2021) found Braess Paradox is still existed on the Braess network for travelers with modern traffic information.

Many studies related to Braess Paradox have been extended to several different system objectives. Nagurney (2000) found that the

* Corresponding author.
E-mail addresses: zhuwenlong@ahu.edu.cn (W. Zhu), zhangjunting2012@126.com (J. Zhang), sqye@ahu.edu.cn (S. Ye), xiangwl@tju.edu.cn (W. Xiang).

https://doi.org/10.1016/j.eswa.2022.118871
Received 2 January 2022; Received in revised form 13 September 2022; Accepted 17 September 2022
Available online 21 September 2022
0957-4174/© 2022 Elsevier Ltd. All rights reserved.

improvements of road network or the reduction of travel demand may result in an increase of the total emissions of network, which are called as Emission Paradoxes. Szeto, Jaber and O'Mahony (2013) explored that the condition of the simultaneous occurrence of Braess and Emission Paradoxes on the Braess networks. Wang and Szeto (2017) examined that the simultaneous and asynchronous occurrences among the excessive noise paradox, the Emission Paradox and the Braess Paradox. Zhang and Waller (2019) showed that after the new link was added to the Braess network with certain travel demand and link performance parameters, both total travel time of network and equity measures related to travel time and energy consumption deteriorated.

Braess Paradox was extended to multi-class travelers with different route choice decision criteria. Considering travelers used perceived travel utility as route choice criteria, Sheffi and Daganzo (1978) found that the Braess Paradox phenomenon based on stochastic traffic assignment model also exists. Zhao, Fu and Wang (2014) used the Logit model to study the Braess Paradox and the robustness under stochastic user equilibrium with elastic demand, and found that Braess Paradox occurs when a new link of the Braess network meets certain condition. Yao and Chen (2014) analyzed the stochastic traffic assignment paradox using the multinomial logit (MNL) model, the multinomial weibit (MNW) model and the hybrid model consisting of MNM and MNW models. Considering bounded rationality assumption, Di et al. (2014) studied the Braess Paradox under boundedly rational user equilibria, and presented the occurrence conditions of the Braess Paradox with respect to indifference band and travel demand. Assume that there are multiple classes of travelers on the road network; Wang et al. (2016) studied the occurrence conditions of the Braess Paradox with mixed equilibrium in the case of non-collaborative Cournot-Nash travelers coexisting with perfectly rational travelers. Considering three different equilibrium conditions, Ameli, Lebacque and Leclercq (2020) designed different banning strategies at some key locations in the road network can improve the network performance.

In reality, many factors, such as travel time, monetary cost, travel time reliability, shorter distance, comfort, safety, fuel consumption and air pollution, influence travelers' route choice behaviors (Abdel-Aty, Kitamura, & Jovanis, 1995; Lam & Small, 2001; Liu, Recker, & Chen, 2004; Ericsson, Larsson, & Brundell-Freij, 2006; Wang, Ehrgott, & Chen, 2014; Kolaka, Feyzioglu, & Noyanb, 2018; Cheng, Zhao, & Li, 2020). Empirical studies have shown that the most important factors are travel time, monetary cost and travel time reliability (Abdel-Aty, Kitamura, & Jovanis, 1995; Lam & Small, 2001; Wang & Ehrgott, 2013). The first two factors (i.e., travel time and monetary cost) have been applied to the relevant studies of traveler's route choice (i.e., Huang & Li, 2007; Cheng, Zhao, & Li, 2020). Considering that all travelers have two objectives (i) minimizing travel time and (ii) minimizing monetary cost, Wang, Raith and Ehrgott (2010) introduced bi-objective user equilibrium, which is more general than user equilibrium with the generalized cost function formulated by the linear combination of time and cost via value of time (also see Wang & Ehrgott, 2013). Wang et al. (2010) have shown that even if the bi-objectives (time-versus-cost) are separable and additive, and either of these objectives does not depend on flow, the bi-objective user equilibrium is not equivalent to the bi-objective version of Beckmann's formulation. Wang and Ehrgott (2013) proved that (i) time surplus maximization bi-objective user equilibrium is equivalent to bi-objective user equilibrium; (ii) bi-objective user equilibrium is a proper generalization of user equilibrium with a generalized cost function. Additionally, Raith et al. (2014) introduced the solution algorithm for the bi-objective traffic assignment or the multi-objective traffic assignment. Moreover, Wang, Ehrgott and Chen (2014) studied the bi-objective user equilibrium model of travel time reliability. Raith (2015) pointed out the solution set of bi-objective user equilibria may be non-convex and studied algorithms for solving bi-objective user equilibrium.

Nevertheless, these studies related to bi-objective user equilibrium focus on the properties and solving algorithms of bi-objective user equilibrium. The occurrence conditions and properties of Braess Paradox under bi-objective user equilibrium have not been investigated in the existing literatures. Furthermore, with the development of Advanced Traveler Information System (ATIS), travelers can get instant and high-quality traffic information involving travel time and monetary cost. Thus, studying the Braess Paradox in the sense of bi-objective user equilibrium can contribute to the improvement of road network planning and the operational efficiency of road traffic systems.

Given the bi-objective decision assumption, how travelers react to the addition of new links on the road network has not been studied. The main purpose of the study is to reveal when the Braess Paradox occurs under the bi-objective route choice behavior on the Braess network. We first define the notion of Braess Paradox and Weak Braess Paradox; formulate the linear performance functions of all links of Braess network. Then, the analytical properties of the bi-objective user equilibrium solutions and the occurrence conditions of Braess Paradox and Weak Braess Paradox in the sense of bi-objective user equilibrium is explored. At last, simulation analysis is tested to present Braess Paradox by adopting the commonly-used simulation data in previous literatures.

The remainder of this paper is organized as follows. In Section 2, related symbols, assumptions, definitions, properties of bi-objective user equilibrium are presented. Section 3 dedicatedly performs Braess Paradox analysis on the Braess network; In Section 4, numerical example is displayed to simulate Braess Paradox phenomenon. Section 5 presents the concluding remarks of this paper.

2. The brief review of Braess Paradox under the bi-objective user equilibrium

A transportation network is considered to be represented by a fully connected directed graph $G[N,\ L]$ where N and L are the set of all nodes and all links, respectively. The notations are listed below.

N: Set of nodes, $|N|$: Number of nodes,
L: Set of links, $|L|$: Number of links,
W: Set of origin–destination (OD) pairs,
d_w: Fixed travel demand between OD pair $w \in W$,
P_w: Set of routes connecting OD pair $w \in W$,
f_p: Route flow on route $p \in P_w, w \in W$,
x_a: Link flow with respect on link $a \in L$,
δ_{ap}: 1 if route p uses link a and 0 otherwise,
t_a: Travel time of traversing link $a \in L$,
c_a: Monetary cost of traversing link $a \in L$,
T_p: Travel time of traversing route $p \in P_w, w \in W$, which satisfies that $T_p = \sum_{a\in L}\delta_{ap}t_a$,
C_p: Monetary cost of traversing route $p \in P_w, w \in W$, which satisfies that $C_p = \sum_{a\in L}\delta_{ap}c_a$,
Ω: Set of feasible route flows, which is expressed as follows:

$$\Omega = \left\{ (f_p)_{p\in P_w,\ w\in W} \,\middle|\, f_p \geqslant 0,\ \sum_{p\in P_w} f_p = d_w,\ p \in P_w,\ w \in W \right\}$$

where $(f_p)_{p\in P_w,\ w\in W}$ denotes route flow vector.

Consider the bi-objective traffic assignment problem, travelers' route choices rely on two independent objectives (i.e., travel time and monetary cost). To introduce the definition of bi-objective user equilibrium, some symbols and order relations need to be used.

For $y^1 = (y_1^1,\ y_2^1),\quad y^2 = (y_1^2,\ y_2^2) \in \mathbb{R}^2$, the following order relations are defined (Raith et al., 2014):

$$y^1 \geqslant y^2 \quad \Leftrightarrow \quad y_k^1 \geqslant y_k^2 \quad \text{for} \quad k = 1, 2,$$

$$y^1 \geqslant y^2 \quad \Leftrightarrow \quad y_k^1 \geqslant y_k^2 \quad \text{for} \quad k = 1, 2; \quad y^1 \neq y^2,$$

$$y^1 > y^2 \quad \Leftrightarrow \quad y_k^1 > y_k^2 \quad \text{for} \quad k = 1, 2.$$

The symbols "$\leqslant, \preccurlyeq, <$" are similarly defined. For convenience, their analytical expressions are omitted.

Firstly, we recall the definition of bi-objective user equilibrium, which is presented by the following Definition 1 (Wang & Ehrgott, 2013).

Definition 1. (*(Bi-objective user equilibrium)*) *Under bi-objective user equilibrium condition traffic arranges itself in such a way that no individual trip maker can improve any of their two objectives without worsening any of the others by unilaterally switching to other route. Mathematically, a network traffic route flow* $\left(f_p\right)_{p\in P_w,\ w\in W} \in \Omega$ *is called bi-objective user equilibrium route flow if and only if for any* $w \in W$ *and any* $p,\ q \in P_w$,

$$\begin{pmatrix} T_p \\ C_p \end{pmatrix} \succcurlyeq \begin{pmatrix} \mathrm{T_q} \\ \mathrm{C_q} \end{pmatrix} \quad \Rightarrow \quad \mathrm{f_p} = 0$$

In particular, bi-objective user equilibrium induces user equilibrium when one of the two objectives is removed. Without loss of generality, we only consider one objective, travel time; the definition of user equilibrium is stated as follows. Under user equilibrium condition traffic arranges itself in such a way that no individual trip maker can improve their travel time by unilaterally switching to other route. Mathematically, a network traffic flow $\left(f_p\right)_{p\in P_w,\ w\in W} \in \Omega$ is called user equilibrium traffic flow if and only if for any $w \in W$ and any $p,\ q \in P_w$,

$$T_p > T_q \quad \Rightarrow \quad f_p = 0$$

Raith (2015) designed the solving method of bi-objective user equilibrium; and pointed out the bi-objective user equilibrium solutions are generally not unique, resulting in different network total travel time. After new links are added to network, the variation of network total travel time is uncertain. However, transportation planners do not want to see that the minimal network total travel time after new links are added exceeds the maximal network total travel time without them. Based on this viewpoint, we introduce the definition of Braess Paradox and Weak Braess Paradox under the bi-objective user equilibrium (See in Di et al., 2014). In the following, for simplicity, Braess Paradox and Weak Braess Paradox are both in the sense of the bi-objective user equilibrium.

Definition 2. (*(Braess Paradox, Weak Braess Paradox)*) (1) Braess Paradox occurs if the minimum of network total travel time after adding new links is higher than the maximum of that without adding new links.

(2) Weak Braess Paradox occurs if the minimum of network total travel time after adding new links is higher than or equal to the maximum of that without adding new links.

It is easy to see that the occurrence of Braess Paradox implies that Weak Braess Paradox must occur, but not vice versa; for example, some newly built roads have never used and network traffic flow patterns have never changed, which is Weak Braess Paradox, is not Braess Paradox.

Replace total travel time with total monetary cost in Definition 2, Braess Paradox and Weak Braess Paradox based on network total monetary cost can be similarly defined, which are called Monetary cost Paradox and Weak Monetary cost Paradox respectively. The simultaneous occurrence of Braess Paradox and Monetary cost Paradox implies that both objectives of the system (total travel time and total monetary cost of network) deteriorate. The conditions for the simultaneous occurrence of these two paradoxes can be obtained by studying these two paradoxes separately. Considering that travel time functions and monetary cost functions have the same functional structure, we next study only the Braess Paradox, the Monetary cost Paradox can be studied in the same method.

3. Braess Paradox analysis on the Braess network

In this section we study the occurrence conditions of Braess Paradox under the bi-objective user equilibrium on the classical Braess network, The Braess network is shown in Fig. 1 with one origin–destination (OD) pair (1, 4). Before a new link is added connecting node 2 and node 3, the Braess network contains four links: link 1–2, link 2–4, link 1–3, link 3–4, and two routes: route 1 (link 1–2 → link 2–4), route 2 (link 1–3 → link 3–4), which is called "two-route scenario" (see in details in Fig. 1 (a)). After a new link is added, compared with the "two-route scenario", the Braess network has one more a link (link 2–3) and one route (route 3: link 1–2 → link 2–3 → link 3–4), which is called "three-route scenario" (see in details in Fig. 1(b)), (Di et al., 2014; Wang et al, 2016).

Before and after the new link 2–3 is added to Braess network, the total travel demand $d_{(1,\ 4)}$ between OD pair (1, 4) is fixed and unchanged, which is abbreviated as d. The travel time functions and monetary cost functions of five links are assumed to be linear functions, which listed as follows:

$$\begin{pmatrix} t_{12} \\ c_{12} \end{pmatrix} = \begin{pmatrix} \beta_1 x_{12} \\ \alpha_2 + \beta_2 x_{12} \end{pmatrix} \tag{1}$$

$$\begin{pmatrix} t_{24} \\ c_{24} \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_2 x_{24} \\ \alpha_1 + \beta_1 x_{24} \end{pmatrix} \tag{2}$$

$$\begin{pmatrix} t_{13} \\ c_{13} \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_2 x_{13} \\ \alpha_1 + \beta_1 x_{13} \end{pmatrix} \tag{3}$$

$$\begin{pmatrix} t_{34} \\ c_{34} \end{pmatrix} = \begin{pmatrix} \beta_1 x_{34} \\ \alpha_2 + \beta_2 x_{34} \end{pmatrix} \tag{4}$$

$$\begin{pmatrix} t_{23} \\ c_{23} \end{pmatrix} = \begin{pmatrix} \alpha_2 + \beta_2 x_{23} \\ 0 \end{pmatrix} \tag{5}$$

where travel time functions of five links follow the linear expressions often used in the previous literatures (see in details in Szeto, Jaber, & O'Mahony, 2013; Di et al., 2014; Wang et al., 2016). Monetary costs are mainly composed of fuel consumption plus congestion charges and subsidies, the existing studies shown that the fuel consumption function was a strictly increasing function of travel time (i.e., Levin & Boyles, 2015). Congestion charges and subsidies are generally constant. For simplicity, monetary cost functions are also assumed to be linear functions, α_1,α_2,β_1, β_2 are four link performance parameters associated with travel time and monetary cost, which satisfy that $\beta_1 > \beta_2 > 0$,$\alpha_1 > \alpha_2 \geqslant 0$. The new link has smaller travel time parameters

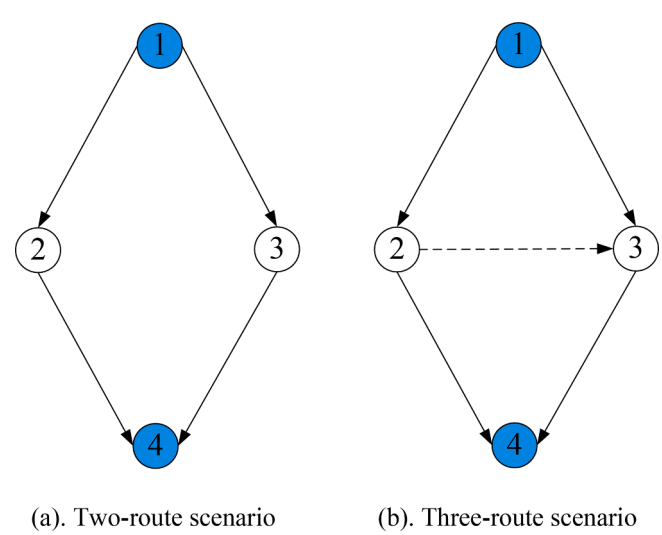

Fig. 1. The Braess network before and after the new link 2–3 is added.

α_2,β_2, and zero monetary cost; these assumptions ensure that the new link 2–3 is attractive.

In the "two-route scenario", the new link 2–3 is not added to the network (Please see in details in Fig. 1(a)), the travel time and monetary cost of two routes are listed below:

$$\begin{pmatrix} T_1 \\ C_1 \end{pmatrix} = \begin{pmatrix} \alpha_1 + (\beta_1 + \beta_2) f_1 \\ (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) f_1 \end{pmatrix} \tag{6}$$

$$\begin{pmatrix} T_2 \\ C_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 + (\beta_1 + \beta_2) f_2 \\ (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) f_2 \end{pmatrix} \tag{7}$$

When the new link 2–3 is added, the network contains two existing routes (route 1 and route 2) and a new route (route 3), and their travel time and monetary cost are calculated as follows:

$$\begin{pmatrix} T_1 \\ C_1 \end{pmatrix} = \begin{pmatrix} \alpha_1 + (\beta_1 + \beta_2) f_1 + \beta_1 f_3 \\ (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) f_1 + \beta_2 f_3 \end{pmatrix} \tag{8}$$

$$\begin{pmatrix} T_2 \\ C_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 + (\beta_1 + \beta_2) f_2 + \beta_1 f_3 \\ (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2) f_2 + \beta_2 f_3 \end{pmatrix} \tag{9}$$

$$\begin{pmatrix} T_3 \\ C_3 \end{pmatrix} = \begin{pmatrix} \alpha_2 + \beta_1 (f_1 + f_2) + (2\beta_1 + \beta_2) f_3 \\ 2\alpha_2 + \beta_2 (f_1 + f_2) + 2\beta_2 f_3 \end{pmatrix} \tag{10}$$

Considering that total travel time of network is the sum of the product of flows and travel time on every route. Thus, we need to determine route flows and travel cost under the bi-objective user equilibrium. Before the new link 2–3 is added, the Braess network structure and the travel time functions and monetary cost functions are symmetric, thus travel costs (travel time and monetary cost) and flows of two routes are equal when the bi-objective user equilibrium condition is reached. Travel costs and route flows of two routes are respectively listed below:

$$\begin{pmatrix} T_1^* \\ C_1^* \end{pmatrix} = \begin{pmatrix} T_2^* \\ C_2^* \end{pmatrix} = \begin{pmatrix} \alpha_1 + \dfrac{(\beta_1 + \beta_2) d}{2} \\ (\alpha_1 + \alpha_2) + \dfrac{(\beta_1 + \beta_2) d}{2} \end{pmatrix} \tag{11}$$

$$f_1^* = f_2^* = \frac{d}{2} \tag{12}$$

We can see from formula (12) that the solution set of the bi-objective user equilibrium is.

$$BUE_2 = \left\{ \left(\frac{d}{2}, \ \frac{d}{2} \right) \right\} \tag{13}$$

Furthermore, the total travel time of the Braess network with 2 routes are computed as follows:

$$T^{2routes} = f_1^* T_1^* + f_2^* T_2^* = \alpha_1 d + \frac{(\beta_1 + \beta_2) d^2}{2} \tag{14}$$

After the new link 2–3 is added to the Braess network, the travel cost structure of the new route 3 is different from the route 1, 2, there are many bi-objective user equilibrium flow patterns, the following Lemma 1 presents the bi-objective user equilibrium flow patterns under 3-route scenario.

Lemma 1. *In the 3-route scenario, the bi-objective user equilibrium route flow patterns are as follows.*

(1). If $\beta_1 > 3\beta_2$, as the level of travel demand increases, the solution set of bi-objective user equilibrium route flow patterns is listed below:

$$BUE_3^{\beta_1 > 3\beta_2} = \begin{cases} \{(m, \ m, \ d - 2m) \mid m = 0\} = \{(0, \ 0, \ d)\}, & 0 < d \leqslant \dfrac{\alpha_1 - \alpha_2}{\beta_1 + \beta_2} \\ \left\{ (m, \ m, \ d - 2m) \mid 0 \leqslant m < \dfrac{(\beta_1 + \beta_2) d - (\alpha_1 - \alpha_2)}{\beta_1 + 3\beta_2} \right\}, & \dfrac{\alpha_1 - \alpha_2}{\beta_1 + \beta_2} < d \leqslant \dfrac{2(\alpha_1 - \alpha_2)}{\beta_1 - \beta_2} \\ \left\{ (m, \ m, \ d - 2m) \mid 0 \leqslant m \leqslant \dfrac{d}{2} \right\}, & \dfrac{2(\alpha_1 - \alpha_2)}{\beta_1 - \beta_2} < d < \dfrac{\alpha_1 - \alpha_2}{\beta_2} \\ \left\{ (m, \ m, \ d - 2m) \mid \dfrac{\beta_2 d - (\alpha_1 - \alpha_2)}{\beta_1 + \beta_2} < m \leqslant \dfrac{d}{2} \right\}, & \dfrac{\alpha_1 - \alpha_2}{\beta_2} \leqslant d < +\infty \end{cases}$$

(2). If $\beta_1 \leqslant 3\beta_2$, as the level of travel demand increases, the solution set of bi-objective user equilibrium route flow patterns is listed below:

$$BUE_3^{\beta_1 \leqslant 3\beta_2} = \begin{cases} \{(m, \ m, \ d - 2m) \mid m = 0\} = \{(0, \ 0, \ d)\}, & 0 < d \leqslant \dfrac{\alpha_1 - \alpha_2}{\beta_1 + \beta_2} \\ \left\{ (m, \ m, \ d - 2m) \mid 0 \leqslant m < \dfrac{(\beta_1 + \beta_2) d - (\alpha_1 - \alpha_2)}{\beta_1 + 3\beta_2} \right\}, & \dfrac{\alpha_1 - \alpha_2}{\beta_1 + \beta_2} < d < \dfrac{\alpha_1 - \alpha_2}{\beta_2} \\ \left\{ (m, \ m, \ d - 2m) \mid \dfrac{\beta_2 d - (\alpha_1 - \alpha_2)}{\beta_1 + \beta_2} < m < \dfrac{(\beta_1 + \beta_2) d - (\alpha_1 - \alpha_2)}{\beta_1 + 3\beta_2} \right\}, & \dfrac{\alpha_1 - \alpha_2}{\beta_2} \leqslant d \leqslant \dfrac{2(\alpha_1 - \alpha_2)}{\beta_1 - \beta_2} \\ \left\{ (m, \ m, \ d - 2m) \mid \dfrac{\beta_2 d - (\alpha_1 - \alpha_2)}{\beta_1 + \beta_2} < m \leqslant \dfrac{d}{2} \right\}, & \dfrac{2(\alpha_1 - \alpha_2)}{\beta_1 - \beta_2} < d < +\infty \end{cases}$$

Furthermore, the total travel time of the Braess network with 3 routes is computed as follows:

$$T^{3routes} = 2(\beta_1 + 3\beta_2)m^2 - 2[2(\beta_1 + \beta_2)d - (\alpha_1 - \alpha_2)]m + [(2\beta_1 + \beta_2)d + \alpha_2]d \tag{15}$$

where m stands for the flows of route 1 under the bi-objective user equilibrium condition.

Proof. See in details in Appendix 1.

In 3-route scenario, given a certain travel demand, the solution set of bi-objective user equilibrium is bounded and convex. These topological properties are helpful for analyzing Braess Paradox under the bi-objective user equilibrium. Besides, link performance parameters β_1, β_2 and travel demand d together affect bi-objective user equilibrium route flow pattern, as shown in Lemma 1, we can see, when travel demand d is relatively small (less than or equal to $\frac{\alpha_1-\alpha_2}{\beta_1+\beta_2}$), there is a unique equilibrium state, at this time, all travelers are willing to switch to the new route (route 3), no travelers remain on the previous routes (route 1 and route 2). When the travel demand d is relatively large (greater than $\frac{\alpha_1-\alpha_2}{\beta_1+\beta_2}$), there are multiple equilibrium states, and as the level of travel demand increases, some travelers will select route 1 and 2; moreover, the lower and upper bounds of the number of travelers on route 1 (or route 2) are monotonically non-decreasing. When the travel demand d exceeds a certain threshold $\frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, no travelers choose the new route, and the number of travelers who choose route 1 and route 2 are the same, both are equal to half of the travel demand, i.e., $\frac{d}{2}$; the route flow pattern is one of multiple bi-objective user equilibrium states.

We next explore the occurrence condition of Braess Paradox under the bi-objective user equilibrium. In the 2-route scenario, the bi-objective user equilibrium route flow pattern is unique. In the 3-route scenario, due to the travel cost characteristics of three routes are asymmetrical, at different travel demand levels, the bi-objective user equilibrium route flow pattern may not be unique. The Braess Paradox occurs if and only if the minimum of total travel time of network in the 3-route scenario is greater than that in the 2-route scenario, i.e., $\min\ T^{3routes} > T^{2routes}$, which demonstrates the addition of a new link makes the Braess network performance worse. In addition, there is another generalized case that deserves our attention, before and after the link 3 is added to the Braess network, the minimum of total travel time of 3-route scenario network greater than or equal to that of 2-route scenario network, i.e., $\min\ T^{3routes} \geqslant T^{2routes}$. In this case, adding new link does not improve network performance, and building new link also consumes human and material resources, which is an unreasonable planning. Based on the above analysis, we study the conditions for the occurrence of Braess Paradox, which are listed as follows.

Theorem 2. *Braess Paradox may occurs in the following cases:*

(1). Braess Paradox occurs if and only if.

$$\frac{2(\alpha_1 - \alpha_2)}{3\beta_1 + \beta_2} < d < \frac{2(\alpha_1 - \alpha_2)}{\beta_1 - \beta_2} \tag{16}$$

(2). Braess Paradox never occurs if and only if.

$$0 < d \leqslant \frac{2(\alpha_1 - \alpha_2)}{3\beta_1 + \beta_2} \tag{17}$$

In particular, if$d = \frac{2(\alpha_1-\alpha_2)}{3\beta_1+\beta_2}$, the new link 2–3 is not used, before and after the new link 2–3 is added, the total travel time is unchanged, Weak Braess Paradox occurs.

(3). If $d > \frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, Weak Braess Paradox occurs and Braess Paradox never occurs. In particular, when the new link 2–3 is used, Braess Paradox occurs; when the new link 2–3 is not used, Braess Paradox does not occur, at this time, the total travel time of network of 3-route scenario and that of 2-route scenario network are the same.

Proof. See in details in Appendix 2.

According to Theorem 2, when travel demand is less than $\frac{2(\alpha_1-\alpha_2)}{3\beta_1+\beta_2}$, the total travel time of network of 3-route scenario smaller than that of 2-route scenario, Braess Paradox never occurs, the increase of the new link can improve the performance of Braess network, and the road planning is reasonable. Using Theorem 2 (2) and (3), we can see that when travel demand is greater than or equal to $\frac{2(\alpha_1-\alpha_2)}{3\beta_1+\beta_2}$, Weak Braess Paradox occurs. that is, the total travel time of augment Braess network is greater than or equal to that of initial Braess network, the performance of Braess network has not improved and road construction costs have been spent, then the addition of the new link is inefficient and this kind of road planning is unreasonable, road planners should avoid redundant or wasteful road construction.

Compared the two Baress Paradox under single objective and bi-objective decision criteria. Under the single objective decision criterion, travelers expect to minimize individual travel costs, and the network traffic flow eventually reaches the user equilibrium state, which is generally unique. Braess Paradox mainly analyzes the system performance before and after the addition of a new link, the occurrence of Braess Paradox depends on link performance parameters of single objective and demand levels. It is different from the previous Braess Paradox analysis under the user equilibrium with the single objective decision criterion, under the bi-objective decision criteria, all travelers have two objectives: minimizing travel time and minimizing monetary cost, and the network traffic flows eventually reach the bi-objective user equilibrium states, which are generally multiple solutions, making the analysis of system performance change more complicated. Braess Paradox mainly analyzes the minimum system performance and the maximum system performance before and after the addition of a new link, the occurrence conditions of Braess Paradox are dependent on link performance parameters of bi-objective and demand levels.

4. Numerical example

In this section, we calculate network flows distribution when Braess Paradox occurs under different travel demand levels, compare the relationship of total travel time of network between 2-route scenario and 3-route scenario. The link performance parameters continue to adopt the commonly used values in the previous research (Nagurney, 2000; Szeto, Jaber, & O'Mahony, 2013; Di et al. 2014; Wang et al. 2016), which is given as follows: $\alpha_1 = 50, \alpha_2 = 10, \beta_1 = 10, \beta_2 = 1$. Because the flow of route 1 and route 2 are the same in the bi-objective user equilibrium states, and the flow of route 3, f_3 can be represented by the flows of route 1 and route 2, f_1 andf_2, i.e., $f_3 = d - f_1 - f_2$. A two-dimensional figure can

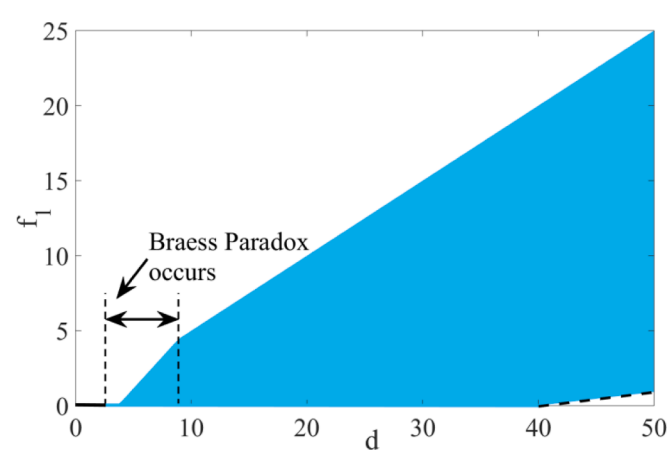

Fig. 2. The flow pattern on route 1 of Braess Paradox occurs in Braess network, where $\beta_1 > 3\beta_2$.

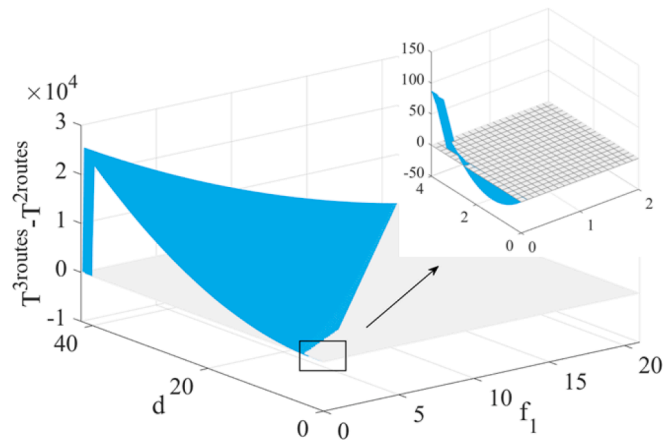

Fig. 3. Total travel time difference of network before and after the new link 2–3 is added under different travel demand.

be used to show the flow patterns of route 1 under the occurrence of Braess Paradox and Weak Braess Paradox. Fig. 2 shows that three kinds of regions:

(1). Braess Paradox occurs, in the blue area pointed by the arrow, where the travel demand $\frac{80}{31} < d \leqslant \frac{80}{9}$.

(2). Braess Paradox never occurs, the solid black line and all blue areas exclude those pointed by the arrow, where the demand $0 < d \leqslant \frac{80}{31}$ and $d > \frac{80}{9}$.

(3). Baress Paradox does not occur and Weak Braess Paradox occurs, The intersection of the black solid line and the blue solid line, plus all blue areas exclude those pointed by the arrow, where the demand $d = \frac{80}{31}$ and $d > \frac{80}{9}$.

We next calculate the difference of the total travel time of network for the Braess network under the 2-route and 3-route scenario, i.e., $T^{3routes} - T^{2routes}$. Given a fixed travel demand, the bi-objective user equilibrium route flow pattern may not be unique, the value of $T^{3routes} - T^{2routes}$ is also affected by bi-objective user equilibrium route flow pattern. Thus, take flow on route 1 and travel demand as independent variables, a three-dimensional figure is adopted to display the value of $T^{3routes} - T^{2routes}$, which is presented by Fig. 3. When the travel demand d is small, the value of $T^{3routes} - T^{2routes}$ is less than 0, the addition of the new link 2–3 can decrease total travel time of network, Braess Paradox and Weak Braess Paradox do not occur. As the travel demand d increases, the value of $T^{3routes} - T^{2routes}$ is always larger than or equal to 0, i.e., plane $T^{3routes} - T^{2routes} = 0$ or above it, where the blue part on the left side of the curve $\{(d, f_1, T^{3routes} - T^{2routes}) \mid d \in (40, 42]\}$ is a collection of their indicator lines, the best scenario is that the total travel time difference of network has not changed, which reflect that building a new link cannot improve total travel time of network, and Weak Braess Paradox occurs.

5. Conclusions

The paper investigated Braess Paradox in the sense of the bi-objective route choice behavior assumption. First, we formulated the link travel time functions and link monetary cost functions to the Braess network, introduced the definitions of Braess Paradox and Weak Braess Paradox under the bi-objective user equilibrium and analyzed the relationship of these two paradoxes. Then, based on Braess network, the solution set of bi-objective user equilibrium was analytically presented, and the occurrence conditions of Braess Paradox and Weak Braess Paradox were explored, which are affected by travel demand and link performance parameters. Lastly, numerical example is used to display total travel time difference of network varies with travel demand before and after the new link is added. Through this study, it is found that the Braess Paradox and Weak Braess Paradox under the bi-objective user equilibrium can occur in Braess network; the addition of new link cannot always improve the efficiency of road traffic systems. Therefore, road planners should sufficiently consider the travel time and monetary cost of new link in practice.

Although the paper is the first attempt to analyze the Braess Paradox under the bi-objective user equilibrium assumption, there are still several limitations. On one hand, from the perspective of travelers, the study assumes that all travelers are rational, homogeneous, and have the same route choice criteria, the heterogeneity and risk attitudes of travelers, and elastic travel demand are not considered in this paper. On the other hand, from the perspective of road network performance, the transportation network uses the classical Braess network, which has a simple structure; and the travel time functions and the monetary cost functions are assumed to be linear functions, this study does not consider the complex transportation network in the real world. In the future study, some issues need to be further explored. First of all, network characteristics of avoiding Braess Paradox when travelers consider bi-objective decision-making criteria needs to be investigated. Secondly, we should explore the occurrence conditions of the Braess Paradox presented in this study under the heterogeneous travelers with different risk attitudes. Thirdly, transportation management schemes including congestion pricing under the framework of bi-objective user equilibrium should be developed to avoid the Braess Paradox. Finally, we expect to apply our Braess Paradox analysis to assist transportation planning department in designing more effective transportation networks in real world and test the system efficiencies of the improved road networks.

CRediT authorship contribution statement

Wenlong Zhu: Conceptualization, Methodology, Software, Writing – original draft. **Junting Zhang:** Conceptualization, Writing – review & editing. **Shunqiang Ye:** Conceptualization, Software, Writing – review & editing. **Wanli Xiang:** Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

The project is supported by Humanity and Social Science Youth Foundation of Ministry of Education of China (18YJC630274, 22YJC630189), National Natural Science Foundation of China (62262037), University Synergy Innovation Program of Anhui Province (GXXT-2021-044), Start-up Funds for Doctors of Anhui University (J01003282).

Appendix 1. Proof of Lemma 1.

In the 3-route scenario, because the route characteristics and travel costs of route 1 and route 2 are symmetric, under the bi-objective user equilibrium is reached, it is easily to obtain that flows of the route 1 and the route 2 are equal. As the level of travel demand varies, the bi-objective user equilibrium route flow patterns may change. According to the relationship between flow patterns and travel costs, the proof divided into three cases:

(1). Only the new route (route 3) carries flows.

In this case, route 1 and 2 do not carry flows, route 3 carries all the traffic flows d, The bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = 0, \quad f_3 = d \tag{18}$$

The corresponding travel time and monetary cost must satisfy the following conditions:

$$\begin{pmatrix} T_1^* \\ C_1^* \end{pmatrix} = \begin{pmatrix} T_2^* \\ C_2^* \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_1 d \\ (\alpha_1 + \alpha_2) + \beta_2 d \end{pmatrix} \geqslant \begin{pmatrix} T_3^* \\ C_3^* \end{pmatrix} = \begin{pmatrix} \alpha_2 + (2\beta_1 + \beta_2)d \\ 2\alpha_2 + 2\beta_2 d \end{pmatrix} \tag{19}$$

or

$$\left(T_1^* - T_3^*\right)\left(C_1^* - C_3^*\right) = [(\alpha_1 - \alpha_2) - (\beta_1 + \beta_2)d][(\alpha_1 - \alpha_2) - \beta_2 d] \langle 0 \tag{20}$$

Thus, we can get the condition for the occurrence of case (1):

$$0 < d < \frac{\alpha_1 - \alpha_2}{\beta_2} \tag{21}$$

The total travel time of network under case (1) is.

$$T_1^{3routes} = \alpha_2 d + (2\beta_1 + \beta_2)d^2 \tag{22}$$

(2). Only the new route (route 3) carries no flows.

In this case, route 3 has no flow, considering the symmetrical characteristics of route 1 and 2, travel demand of network d is equally assigned to these two routes. Thus, the bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = \frac{d}{2}, \quad f_3 = 0 \tag{23}$$

The corresponding travel time and monetary cost can be described as follows:

$$\begin{pmatrix} T_1^* \\ C_1^* \end{pmatrix} = \begin{pmatrix} T_2^* \\ C_2^* \end{pmatrix} = \begin{pmatrix} \alpha_1 + \dfrac{(\beta_1 + \beta_2)d}{2} \\ (\alpha_1 + \alpha_2) + \dfrac{(\beta_1 + \beta_2)d}{2} \end{pmatrix} \leqslant \begin{pmatrix} T_3^* \\ C_3^* \end{pmatrix} = \begin{pmatrix} \alpha_2 + \beta_1 d \\ 2\alpha_2 + \beta_2 d \end{pmatrix} \tag{24}$$

or

$$\left(T_1^* - T_3^*\right)\left(C_1^* - C_3^*\right) = \left[(\alpha_1 - \alpha_2) - \frac{(\beta_1 - \beta_2)d}{2}\right]\left[(\alpha_1 - \alpha_2) + \frac{(\beta_1 - \beta_2)d}{2}\right] \Big\langle 0 \tag{25}$$

Thus, it is easily to obtain that the condition for the occurrence of case (2):

$$d > \frac{2(\alpha_1 - \alpha_2)}{\beta_1 - \beta_2} \tag{26}$$

The total travel time of network under case (2) is.

$$T_2^{3routes} = \alpha_1 d + \frac{(\beta_1 + \beta_2)d^2}{2} \tag{27}$$

(3). All routes carry positive flows.

In this case, three routes are used. Due to the symmetrical characteristics of route 1 and 2, the two routes flows are the same, assuming m,$0 < m < \frac{d}{2}$. Thus, the bi-objective user equilibrium flow pattern as:

$$f_1 = f_2 = m, \quad f_3 = d - 2m, \quad 0 < m < \frac{d}{2} \tag{28}$$

The corresponding travel time and monetary cost satisfy the following conditions:

$$\begin{aligned} \begin{pmatrix} T_1^* \\ C_1^* \end{pmatrix} &= \begin{pmatrix} T_2^* \\ C_2^* \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_1 d - (\beta_1 - \beta_2)m \\ (\alpha_1 + \alpha_2) + \beta_2 d + (\beta_1 - \beta_2)m \end{pmatrix} \\ &= \begin{pmatrix} T_3^* \\ C_3^* \end{pmatrix} = \begin{pmatrix} \alpha_2 + (2\beta_1 + \beta_2)d - 2(\beta_1 + \beta_2)m \\ 2\alpha_2 + 2\beta_2 d - 2\beta_2 m \end{pmatrix} \end{aligned} \tag{29}$$

or

$$\left(T_1^* - T_3^*\right)\left(C_1^* - C_3^*\right) = (\beta_1 + \beta_2)(\beta_1 + 3\beta_2)\left[m - \frac{(\beta_1 + \beta_2)d - (\alpha_1 - \alpha_2)}{\beta_1 + 3\beta_2}\right]\left[m - \frac{\beta_2 d - (\alpha_1 - \alpha_2)}{\beta_1 + \beta_2}\right] \Big\langle 0 \tag{30}$$

Subsequently, by solving the equation (29) and inequality (30), we can obtain the conditions that the route flow reaches the bi-objective user equilibrium state, which is affected by link performance parameters β_1 andβ_2, and the following proof needs to be discussed is divided into two cases:

(I). If $\beta_1 > 3\beta_2$, as the travel demand d varies, there are three kinds of equilibrium route flow patterns:

(I_1). When $\frac{\alpha_1-\alpha_2}{\beta_1+\beta_2} < d \leqslant \frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, the bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = m,\ f_3 = d - 2m, \quad 0 < m < \frac{(\beta_1+\beta_2)d - (\alpha_1-\alpha_2)}{\beta_1+3\beta_2} \tag{31}$$

(I_2). When $\frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2} < d < \frac{\alpha_1-\alpha_2}{\beta_2}$, the bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = m,\ f_3 = d - 2m, \quad 0 < m < \frac{d}{2} \tag{32}$$

(I_3). When $\frac{\alpha_1-\alpha_2}{\beta_2} \leqslant d < +\infty$, the bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = m,\ f_3 = d - 2m, \quad \frac{\beta_2 d - (\alpha_1-\alpha_2)}{\beta_1+\beta_2} < m < \frac{d}{2} \tag{33}$$

(II). If $\beta_1 \leqslant 3\beta_2$, as the travel demand d varies, there are three kinds of equilibrium route flow patterns:

(II_1). When $\frac{\alpha_1-\alpha_2}{\beta_1+\beta_2} < d < \frac{\alpha_1-\alpha_2}{\beta_2}$, the bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = m,\ f_3 = d - 2m, \quad 0 < m < \frac{(\beta_1+\beta_2)d - (\alpha_1-\alpha_2)}{\beta_1+3\beta_2} \tag{34}$$

(II_2). When $\frac{\alpha_1-\alpha_2}{\beta_2} \leqslant d \leqslant \frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, the bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = m,\ f_3 = d - 2m, \quad \frac{\beta_2 d - (\alpha_1-\alpha_2)}{\beta_1+\beta_2} < m < \frac{(\beta_1+\beta_2)d - (\alpha_1-\alpha_2)}{\beta_1+3\beta_2} \tag{35}$$

(II_3). When $\frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2} < d < +\infty$, the bi-objective user equilibrium route flow pattern as:

$$f_1 = f_2 = m,\ f_3 = d - 2m, \quad \frac{\beta_2 d - (\alpha_1-\alpha_2)}{\beta_1+\beta_2} < m < \frac{d}{2} \tag{36}$$

Furthermore, the total travel time of network under case (3) is computed as follows:

$$T^{3routes} = 2(\beta_1+3\beta_2)m^2 - 2[2(\beta_1+\beta_2)d - (\alpha_1-\alpha_2)]m + [(2\beta_1+\beta_2)d + \alpha_2]d \tag{37}$$

Combining formulas (18), (21), and (31)-(33), the bi-objective user equilibria $BUE_3^{\beta_1>3\beta_2}$ is obtained; by using formulas (23), (26), and (34)-(36), the bi-objective user equilibria $BUE_3^{\beta_1\leqslant 3\beta_2}$ is obtained.

We next calculate total travel time of network under the condition that the route flow located in the solution set of bi-objective user equilibrium $BUE_3^{\beta_1>3\beta_2}$ or $BUE_3^{\beta_1\leqslant 3\beta_2}$. In formula (15), set m equal 0 and $\frac{d}{2}$, which are equivalent to formulas (22) and (27), respectively. Therefore, the total travel time of network under three different cases can be integrated into the same formula (formula (15)) to express, and the proof is finished.

Appendix 2. Proof of Theorem 2.

In 2-route scenario, the bi-objective user equilibrium solution is unique, the total travel time of network is a certain value, and its analytical expression is given by formula (14). In 3-route scenario, it can be seen from Lemma 1 that there may be multiple bi-objective user equilibrium solutions, the total travel time of network changes with the bi-objective user equilibrium flows, the calculation method is given by formula (15). According to the definition, the Braess Paradox occurs if and only if $(\min\ T^{3routes}) \rangle T^{2routes}$, we next need to calculate the value of $T^{3routes} - T^{2routes}$, which is expressed as follows:

$$T^{3routes} - T^{2routes} = 2(\beta_1+3\beta_2)\left(m - \frac{d}{2}\right)\left(m - \frac{(3\beta_1+\beta_2)d - 2(\alpha_1-\alpha_2)}{2(\beta_1+3\beta_2)}\right) \tag{38}$$

According to Lemma 1, link performance parameters $\beta_1 > 3\beta_2$ or $\beta_1 \leqslant 3\beta_2$, the bi-objective user equilibrium route flow patterns are different, then the following proof is divided into two cases:

Case (I). When link performance parameters satisfy condition $\beta_1 > 3\beta_2$.

(I_a). If $0 < d \leqslant \frac{\alpha_1-\alpha_2}{\beta_1+\beta_2}$, no traveler selects route 1 and route 2, i.e., $m = 0$, then.

$$T^{3routes} - T^{2routes} = \frac{(3\beta_1+\beta_2)d^2 - 2(\alpha_1-\alpha_2)d}{2} \tag{39}$$

When $\frac{2(\alpha_1-\alpha_2)}{3\beta_1+\beta_2} < d \leqslant \frac{\alpha_1-\alpha_2}{\beta_1+\beta_2}$, we obtain $T^{3routes} - T^{2routes} > 0$, thus the Braess Paradox occurs. When $0 < d < \frac{2(\alpha_1-\alpha_2)}{3\beta_1+\beta_2}$, $T^{3routes} - T^{2routes} < 0$, the Braess Paradox never occur.

(I_b). If $\frac{\alpha_1-\alpha_2}{\beta_1+\beta_2} < d \leqslant \frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, by Lemma 1, the flow m of route 1 satisfies the condition $0 \leqslant m < \frac{(\beta_1+\beta_2)d-(\alpha_1-\alpha_2)}{\beta_1+3\beta_2}$, combined with formula (38), we can verify that $T^{3routes} - T^{2routes}$ is always greater than 0, thus Braess Paradox always occurs.

(I_c). If $d > \frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, it is easily to verify that $T^{3routes} - T^{2routes}$ is always greater than or equal to 0. Moreover, $T^{3routes} - T^{2routes} = 0$ if and only if $m = \frac{d}{2}$. Thus either Braess Paradox occurs or the new link 2–3 is not used.

Case (II). When link performance parameters satisfy condition $\beta_1 \leqslant 3\beta_2$.

(II_a). If $0 < d \leqslant \frac{\alpha_1-\alpha_2}{\beta_1+\beta_2}$, the proof is the same as (I_a).

(II_b). If $\frac{\alpha_1-\alpha_2}{\beta_1+\beta_2} < d \leqslant \frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, by Lemma 1, the flow m of route 1 always does not exceed $\frac{(\beta_1+\beta_2)d-(\alpha_1-\alpha_2)}{\beta_1+3\beta_2}$, which implies that $T^{3routes} - T^{2routes}$ is always greater than 0, thus Braess Paradox occurs.

($\mathrm{II_c}$). If $d > \frac{2(\alpha_1-\alpha_2)}{\beta_1-\beta_2}$, by using Lemma 1 again, under the bi-objective user equilibrium states, the flow m of route 1 is always less than or equal to half of travel demand, i.e., $m \leqslant \frac{d}{2}$ we can verify that $T^{3routes} - T^{2routes}$ is always greater than or equal to 0. Moreover, $T^{3routes} = T^{2routes}$ if and only if $m = \frac{d}{2}$. Thus either Braess Paradox occurs or the new link 2–3 is not used.

In summary, combining the above two cases: case (I) and case (II), the proof of Theorem 2 is completed.

References

Abdel-Aty, M. A., Kitamura, R., & Jovanis, P. P. (1995). Investigating effect of travel time variability on route choice using repeated-measurement stated preference data. *Transportation Research Record, 1493*, 39–45.

Ameli, M., Lebacque, J. P., & Leclercq, L. (2020). Improving traffic network performance with road banning strategy: A simulation approach comparing user equilibrium and system optimum. *Simulation Modelling Practice and Theory, 99*, Article 101995.

Bagloee, S. A., Ceder, A., Tavana, M., & Bozic, C. (2014). A heuristic methodology to tackle the Braess Paradox detecting problem tailored for real road networks. *Transportmetrica A: Transport Science, 10*(5), 437–456.

Bittihn, S., & Schadschneider, A. (2020). Braess' paradox in the age of traffic information. *Journal of Statistical Mechanics: Theory and Experiment, 2020*, Article 033401.

Bittihn, S., & Schadschneider, A. (2021). The effect of modern traffic information on Braess' paradox. *Physica A: Statistical Mechanics and its Applications, 571*(2021), Article 125829.

Braess, D. (1968). Über ein paradoxon aus der Verkehrsplanung. *Unternehmensforschung, 12*, 258–268.

Braess, D., Nagurney, A., & Wakolbingger, T. (2005). On a paradox of traffic planning. *Transportation Science, 39*(4), 446–450.

Cheng, Z., Zhao, L., & Li, H. (2020). A transportation network paradox: Considering of travel time and health damage due pollution. *Sustainability, 12*.

Çömez-Dolgan, N., & Fescioglu-Unver, N. (2015). Managing transshipments in a multi-retailer system with approximate policies. *Journal of the Operational Research Society, 66*, 947–964.

Di, X., He, X., Guo, X., & Liu, H. X. (2014). Braess Paradox under the boundedly rational user equilibria. *Transportation Research Part B, 67*, 86–108.

Ericsson, E., Larsson, H., & Brundell-Freij, K. (2006). Optimizing route choice for lowest fuel consumption-Potential effects of a new driver support tool. *Transportation Research Part C, 14*(6), 369–383.

Huang, H. J., & Li, Z. C. (2007). A multiclass, multicriteria logit-based traffic equilibrium assignment model under ATIS. *European Journal of Operational Research, 176*(3), 1464–1477.

Kolaka, O. I., Feyzioglu, O., & Noyanb, N. (2018). Bi-level multi-objective traffic network optimization with sustainability perspective. *Expert Systems With Applications, 104*, 294–306.

Lam, T. C., & Small, K. A. (2001). The value of time and reliability: Measurement from a value pricing experiment. *Transportation Research Part E, 37*(2–3), 231–251.

Levin, M. W., & Boyles, S. D. (2015). Effects of autonomous vehicle ownership on trip, mode, and route choice. *Transportation Research Record: Journal of the Transportation Research Board, 2015*(2493), 29–38.

Liu, H., Recker, W., & Chen, A. (2004). Uncovering the contribution of travel time reliability to dynamic route choice using real-time loop data. *Transportation Research Part A, 38*(6), 435–453.

Liu, S., Guo, L., Easa, S. M., Yan, H., Wei, H., & Tang, Y. (2020). Experimental study of day-to-day route-choice behavior: Evaluating the effect of ATIS market penetration. *Journal of Advanced Transportation, 2020*(3). https://doi.org/10.1155/2020/8393724

Ma, C., Cai, Q., Alam, S., Sridhar, B., & Duong, V. N. (2019). Airway network management using Braess's Paradox. *Transportation Research Part C, 105*, 565–579.

Morgan, J., Orzen, H., & Sefton, M. (2009). Network architecture and traffic flows: Experiments on the Pigou-Knight-Downs and Braess Paradoxes. *Games and Economic Behavior, 66*, 348–372.

Murchland, J. D. (1970). Braess paradox of traffic flow. *Transportation Research, 4*(4), 391–394.

Nagurney, A. (2000). Congested urban transportation networks and emission paradoxes. *Transportation Research Part D, 5*, 145–151.

Nagurney, A. (2010). The negation of the Braess paradox as demand increases: The wisdom of crowds in transportation networks. *Europhysics Letters, 91*(4), 48002.

Nagurney, A., Parkes, D., & Daniele, P. (2007). The Internet, evolutionary variational inequalities, and the time-dependent Braess paradox. *Computational Management Science, 4*(4), 355–375.

Passacantando, M., Gnecco, G., Hadas, Y., & Sanguineti, M. (2021a). Braess' paradox: A cooperative game-theoretic point of view. *Networks, 78*(3), 264–283.

Passacantando, M., Gnecco, G., Hadas, Y., & Sanguineti, M. (2021b). On Braess' paradox and average quality of service in transportation network cooperative games. *Optimization and Decision Science, AIRO Springer Series, 7*, 27–37.

Raith, A. (2015). *Algorithms for multi-objective network equilibrium problems*. University of Auckland. Dissertation.

Raith, A., Wang, J. Y. T., Ehrgott, M., & Mitchell, S. A. (2014). Solving multi-objective traffic assignment. *Annals of Operations Research, 222*, 483–516.

Braess Paradox in the laboratory: experimental study of route choice in traffic networks with asymmetric costs (vol. 21), (2008). https://doi.org/10.1007/978-0-387-77131-1_13

Sheffi, Y., & Daganzo, C. F. (1978). Another "paradox" of traffic flow. *Transportation Research, 12*, 43–46.

Steinberg, R., & Zangwill, W. I. (1983). The prevalence of Braess' paradox. *Transportation Science, 17*(3), 301–318.

Szeto, W. Y., Jaber, X. Q., & O'Mahony, M. (2013). On the existence of the simultaneous occurrence of Braess and emission paradoxes. *Transportation Research Board Annual Meeting Proceedings*.

Valiant, G., & Roughgarden, T. (2006). Braess's paradox in large random graphs. Proceedings of the 7th ACM conference on Electronic commerce (EC'06), 296-305.

Wang, Judith Y.T., Raith, A., & Ehrgott, M. (2010). Tolling Analysis with Bi-objective Traffic Assignment. Pages 117–129 of: Ehrgott, M., Naujoks, B., Stewart, T., Wallenius, J. (eds), Multiple Criteria Decision Making for Sustainable Energy and Transportation Systems, Springer Verlag, Berlin.

Wang, J. Y. T., & Ehrgott, M. (2013). Modelling route choice behaviour in a tolled road network with a time surplus maximisation bi-objective user equilibrium model. *Transportation Research Part B, 57*, 342–360.

Wang, J. Y. T., Ehrgott, M., & Chen, A. (2014). A bi-objective user equilibrium model of travel time reliability in a road network. *Transportation Research Part B, 66*, 4–15.

Wang, W., Wang, D. Z. W., Sun, H., Feng, Z., & Wu, J. (2016). Braess Paradox of traffic networks with mixed equilibrium behaviors. *Transportation Research Part E, 93*, 95–114.

Wang, Y., & Szeto, W. Y. (2017). Excessive noise paradoxes in urban transportation networks. *Transportmetrica A: Transport Science, 13*(3), 195–221.

Yao, J., & Chen, A. (2014). An analysis of logit and weibit route choices in stochastic assignment paradox. *Transportation Research Part B, 69*, 31–49.

Zhang, X., & Waller, S. T. (2019). Implications of link-based equity objectives on transportation network design problem. *Transportation, 46*, 1559–1589.

Zhao, C., Fu, B., & Wang, T. (2014). Braess Paradox and robustness of traffic network under stochastic user equilibrium. *Transportation Research Part E, 61*, 135–141.